Section 51. $\qquad\qquad\qquad$ mathscinec.

#4. $irr(i, \mathbb{Q}) = x^2+1 \ (= (x+i)(x-i) \text{ in } \overline{\mathbb{Q}})$

$irr(\sqrt[3]{2}, \mathbb{Q}) = x^3-2 \ \left(= \prod_{k=0,1,2} (x - \sqrt[3]{2}\cdot \exp(2\pi i \cdot \tfrac{k}{3})) \text{ in } \overline{\mathbb{Q}} \right)$

$\mathbb{Q}(\sqrt[3]{2}, i)$)

Take $a \in \mathbb{Q}$ satisfying

$$a \neq \frac{\sqrt[3]{2}}{2i}\cdot\left(-1 + \exp(2\pi i\cdot \tfrac{k}{3})\right) \ (k=1,2).$$

$$= \frac{3\sqrt{2}}{2i}\left(-\frac{3}{2} + \frac{\sqrt{3}}{2}i\right) \text{ or } \frac{3\sqrt{2}}{2i}\left(-\frac{3}{2} - \frac{\sqrt{3}}{2}i\right)$$

$$\boxed{\frac{\sqrt{2}, \sqrt{3}}{\sqrt{2}+\sqrt{3}}}$$

Let $a=1$, and $\alpha = \sqrt[3]{2} + i$. Denote $\sqrt[3]{2} = \tau$. ($\tau^3 = 2$, clearly)

$\alpha^2 = \tau^2 - 1 + 2i\tau = (-1, 0, 1, 0, 2, 0)$

$\alpha^3 = 2 - i + 3\tau^2 i - 3\tau = (2, -3, 0, -1, 0, 3)$ $\qquad$ * $\beta = (1, \tau, \tau^2, i, i\tau, i\tau^2)$

$\alpha^4 = 2\tau - i\tau + 6i - 3\tau^2 + 2i + 1 - 3\tau^2 - 3\tau i = (1, 2, -6, 8, -4, 0)$

$\alpha^5 = (-8, 4, 0, 1, 2, -6) + (-12, 1, 2, 0, 8, -4) = (-20, 5, 2, 1, 10, -10)$

(Since $[\mathbb{Q}(\sqrt[3]{2}, i) : \mathbb{Q}] = 6$, α^n $(n \geq 6)$ are meaningless).

The set $\{\alpha^n \mid 0 \leq n \leq 5\}$ is linearly independent, therefore, we can find two polynomials in α as

$$i = (0, 0, 0, 1, 0, 0) = \sum_{n=0}^{5} a_n \alpha^n \qquad \text{where } a_n, b_n \in F.$$

$$\sqrt[3]{2} = (0, 1, 0, 0, 0, 0) = \sum_{n=0}^{5} b_n \alpha^n$$

Take $A = \left(1 \ [\alpha]_\beta \ [\alpha^2]_\beta \cdots [\alpha^5]_\beta\right) \in M_{6\times 6}(F)$

$$A\begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} = i = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \iff \begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} = A^{-1}\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \left(\text{Specifically, } (a_0, \cdots, a_5) = \frac{1}{22}\cdot(91, 100, -18, 40, -9, 12)\right)$$

and

$$A\begin{pmatrix} b_0 \\ \vdots \\ b_5 \end{pmatrix} = \tau = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \iff \begin{pmatrix} b_0 \\ \vdots \\ b_5 \end{pmatrix} = A^{-1}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Finally, calculating A^{-1}, we obtain the result.

Alternatively, take $\alpha = \sqrt[3]{2} i$. Then, $\sqrt[3]{2} = \alpha^4 \cdot \frac{1}{2}$ and $i = -2\alpha^3$, thus

$F(\alpha) = F(\sqrt[3]{2}, i)$.

#11. Algebraic extension of perfect is perfect.

$$\begin{array}{c} K \\ | \\ E \\ | \\ F \end{array}$$

proof. Let F be a perfect field and E/F be an algebraic extension.
Let K/E be a finite extension of E.
Suppose that K is not separable over E. Then,

$$\{K:E\} < [K:E].$$

This implies that for some $\alpha \in K$, $irr(\alpha, E)$ has common multiplicity larger than 1.
Meanwhile, since K/E finite and E/F algebraic, $\alpha \in K$ is algebraic over F.
But, since F is a perfect field, $irr(\alpha, F)$ is separable over F.
But!! $irr(\alpha, E)$ divides $irr(\alpha, F)$, it is contradiction.

#12. Let $F \leq E \leq K$, E/F algebraic, K/E algebraic.
Let $\alpha \in K$ be given. Since α is algebraic over E and E/F algebraic,
α is algebraic over F. Therefore, $irr(\alpha, F)$ exists.
By the assumption, $irr(\alpha, E)$ have common multiplicity as 1. Denote

$$irr(\alpha, E) = \sum a_n x^n \quad \text{where } a_n \in E.$$

Since $irr(\alpha, E)$ is contained in $F(a_0, \cdots, a_{\deg_E \alpha})$ with multiplicity 1,
α is separable over $F(a_0, \cdots, a_{\deg_E \alpha})$, which implies that
α is separable over F. ($F(a_0, \cdots, a_{\deg_E \alpha})$ is separable, by the assumption)

#13. Let $F_s = \{a \in E \mid a \text{ is separable over } F\}$

Clearly $F \leq F_s$. Let $\alpha, \beta \in F_s$ be given.
Since $F(\alpha \pm \beta) \subseteq F(\alpha, \beta)$ and $F(\alpha, \beta)$ is separable over F, $F(\alpha \pm \beta)$ is separable.
Similarly $F(\alpha\beta)$ and $F(\alpha/\beta)$ $(\beta \neq 0)$ too.

#14.

a). for any $\alpha \in E$, $\sigma_p^n(\alpha) = \alpha^{p^n} = \alpha$, because the finite field of order p^n is set of zeros of $x^{p^n} - x$.

If order of σ_p is smaller than n, then it is contradiction to the above fact.

b). Since E is finite field, $E/\mathbb{Z}_p$ separable and simple extension.

And E is splitting field of $\mathbb{Z}_p$, because it is generated by all roots of $x^{p^n-1} - 1$.

Now, $|G(E/\mathbb{Z}_p)| = \{E : \mathbb{Z}_p\} = [E : \mathbb{Z}_p] = n$ with

$\sigma_p \in G(E/\mathbb{Z}_p)$ has order of n,

$G(E/\mathbb{Z}_p) = \langle \sigma_p \rangle$.

$\sqrt[3]{2}$, $\sqrt[3]{2} \cdot \frac{-1+i\sqrt{3}}{2}$, $\sqrt[3]{2} \cdot \frac{-1-i\sqrt{3}}{2}$

「Section 53,

#4. Clearly, $[K : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 2$. Thus $|\lambda(\mathbb{Q}(\sqrt{2}, \sqrt{3}))| = 2$.

#7. similarly, $[K : \mathbb{Q}(\sqrt{2}+\sqrt{6})] = 2$. gives that $|\lambda(\mathbb{Q}(\sqrt{3}, \sqrt{3}))| = 2$.

(Since $[\mathbb{Q}(\sqrt{2}+\sqrt{6}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}+\sqrt{6}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 4$,
$[K : \mathbb{Q}(\sqrt{2}+\sqrt{6})] = 2$.)

#11. Find $Gal(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})$. Denote $\tau_k = \sqrt[3]{2} \cdot \exp(2\pi i \cdot \frac{k}{3})$ $k = 1, 2$

a). σ_0 = identity. σ_1 $\sqrt[3]{2}$ maps to τ_1 and $i\sqrt{3}$ maps to $i\sqrt{3}$ 3

σ_2 〃 to τ_2 and 〃 2

σ_3 〃 to $\sqrt[3]{2}$ and $i\sqrt{3}$ maps to $-i\sqrt{3}$ 2

σ_4 〃 to τ_1 and 〃 3

σ_5 〃 to τ_2 and 〃 2

b). By the Cauchy's Theorem, group of order 6 is $\mathbb{Z}_6$ if it is Abelian, and S_3 if it is non-Abelian.

Enough to show $G(K/\mathbb{Q})$ is not Abelian. Observe that

$(\sigma_1 \circ \sigma_3)(\sqrt[3]{2}) = \sigma_1(\sqrt[3]{2}) = \tau_1$ but

$(\sigma_3 \circ \sigma_1)(\sqrt[3]{2}) = \sigma_3(\tau_1) = \sqrt[3]{2} \cdot (-1 - i\sqrt{3})/2 \neq \tau_1$. Thus $G(K/\mathbb{Q})$ is not Abelian.

$\sigma_1^2(\sqrt[3]{2}) = \sigma_1(\tau_1) = \tau_1$.

$\sigma_5^2(\sqrt[3]{2}) = \sigma_5(\tau_2) =$

c)

$$
\begin{tikzcd}[column sep=small]
 & & K \arrow[dll, dash, "2"'] \arrow[dl, dash] \arrow[d, dash] \arrow[drr, dash, "3"] & & \\
\mathbb{Q}(\sqrt[3]{2}) \arrow[drr, dash] & \mathbb{Q}(\tau_1) \arrow[dr, dash] & \mathbb{Q}(\tau_2) \arrow[d, dash] & & \mathbb{Q}(i\sqrt{3}) \arrow[dll, dash] \\
 & & \mathbb{Q} & &
\end{tikzcd}
$$

$$
\begin{tikzcd}[column sep=small]
 & & \{\sigma_0\} \arrow[dll, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\{\sigma_0, \sigma_1, \sigma_4\} \arrow[drr, dash] & & \{\sigma_0, \sigma_2\} \arrow[d, dash] & \{\sigma_0, \sigma_3\} \arrow[dl, dash] & \{\sigma_0, \sigma_5\} \arrow[dll, dash] \\
 & & G(K/\mathbb{Q}) & &
\end{tikzcd}
$$

#6. Since $G(K/E) \trianglelefteq G(K/F)$,

$$G(E/F) \cong G(K/F) / G(K/E)$$

and $G(K/F)$ is abelian give the $G(E/F)$ is abelian.

$G(K/E)$ is abelian, because it is subgroup of abelian.

⌈Thm. Let K be a finite normal extension of F, and $F \leq E \leq K$. Then K is a finite normal extension of E and

$$\begin{array}{c} K \\ | \\ E \\ | \\ F \end{array}$$

$$Aut(K/E) \leq Aut(K/F).$$

Moreover, two Automorphisms σ and τ in $Aut(K/F)$ are satisfied $\sigma|_E = \tau|_E$

if and only if

$$\sigma Aut(K/E) = \tau Aut(K/E)$$

Proof. Above two assertions are trivial. Let $\sigma, \tau \in Aut(K/F)$ be given.

$$\sigma Aut(K/E) = \tau Aut(K/E) \iff \tau^{-1}\sigma = \mu \text{ for some } \mu \in Aut(K/E)$$
$$\iff \sigma|_E = \tau|_E$$

⌈Def. If K is a finite Normal Extension of F, then $Aut(K/F)$ is called

Galois Group of K over F.

⌈Galois Theory.

Let K/F be a finite normal extension.

$$\lambda : \{E \mid F \leq E \leq K\} \to Gal(K/F) : E \mapsto Gal(K/E)$$

For any $F \leq E \leq K$, $[K:E] = |Gal(K/E)|$,

$[E:F] = (Gal(K/F) : Gal(K/E))$

E is Normal of F $\iff$ $G(K/E)$ is Normal subgroup of $G(K/F)$.

If E is Normal of F, then

$$G(E/F) \cong G(K/F)/G(K/E)$$

$\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$

$\sqrt{2} \cdot \sqrt{3}$

$$\begin{tikzcd}[column sep=small]
& \mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2}+\sqrt{3}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash] & \mathbb{Q}(\sqrt{3}) \arrow[d, dash] & \mathbb{Q}(\sqrt{6}) \arrow[dl, dash] \\
& \mathbb{Q} &
\end{tikzcd}
\iff
\begin{tikzcd}[column sep=small]
& K \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle a \rangle \arrow[dr, dash] & \langle b \rangle \arrow[d, dash] & \langle ab \rangle \arrow[dl, dash] \\
& \{1\} &
\end{tikzcd}$$

$\mathrm{irr}(\sqrt{2}, \mathbb{Q})$ $\quad$ $\mathrm{irr}(\sqrt{3}, \mathbb{Q})$

$\alpha = \sqrt{2}+\sqrt{3}$

$\alpha^2 = 5 + 2\sqrt{6}$

$\alpha^3 = 5\sqrt{2} + 4\sqrt{3} + 5\sqrt{3} + 6\sqrt{2} = 11\sqrt{2} + 9\sqrt{3}$

$(\alpha^3 - 11\alpha) \cdot -\frac{1}{2} = \sqrt{3}$

$(\alpha^3 - 9\alpha) \cdot \frac{1}{2} = \sqrt{2}$

$(a_1 + \cdots + a_m)^n$

$= (a_1 + (a_2 + \cdots + a_m))^n$

$= \sum \binom{n}{k_1} a_1^{n-k_1} \cdot (\quad)^{k_1}$